

Talend Configuration Guide

Document Information

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Important: This guide provides organization-level configuration guidance aligned with the governance framework in this repository. Always follow Talend's official documentation and your license/support agreement for product-specific instructions, as features and UI vary by product (e.g., Talend Data Fabric, Open Studio) and version.

Purpose

This guide maps [Organization Name]'s data governance framework onto a Talend implementation, covering integration job standards, metadata/lineage capture, and data quality rule execution, so the tool reinforces the governance program defined in this repository.

1. Pre-Implementation Checklist

- [] Data Governance Charter and Classification Scheme finalized
- [] Pilot domain and source systems identified (per Implementation Roadmap)
- [] Talend Studio/project workspace provisioned with environment separation (Dev/Test/Prod)
- [] Source/target connection metadata documented per Data_Catalog_Setup_Scripts.sql
`data\_sources` table

2. Project and Job Organization Standards

- [] Organize Talend projects/folders by business domain, mirroring `business\_domains` in the catalog schema
- [] Adopt a consistent job naming convention: `[domain]\_[source]\_[target]\_[frequency]` (e.g., `finance\_erp\_to\_dw\_daily`)
- [] Tag jobs with metadata (owner, criticality, classification of data processed) for traceability

3. Metadata and Lineage Capture

- ☐ Enable Talend's built-in lineage/impact analysis features (where licensed) to auto-document source-to-target flows
- ☐ For jobs not covered by native lineage tooling, run completed job SQL/tMap logic through `Data_Lineage_Discovery_Tools.py` to generate candidate lineage records
- ☐ Register all production jobs and their source/target assets in the data catalog, with Technical Data Steward review before marking lineage "Verified"

4. Data Quality Integration

- ☐ Implement Talend Data Quality components (tDataQuality, tSchemaComplianceCheck, etc.) aligned to the rule dimensions in the Data Quality Rules Library
- ☐ Route quality check failures to a centralized logging table matching the ``data_quality_results`` schema
- ☐ Configure job-level alerts for Critical Data Element quality failures, routed to the Data Issue Resolution Process

5. Error Handling and Logging Standards

- ☐ Standardize job error handling: all jobs must log failures to a central log (matching ``data_issues`` schema) rather than failing silently
- ☐ Configure retry logic for transient failures (e.g., network timeouts) before escalating to manual intervention
- ☐ Configure notifications to the responsible Technical Data Steward on job failure for Critical Data Element pipelines

6. Change Management

- ☐ All production job changes require code review and Technical Data Steward sign-off for jobs touching Critical Data Elements
- ☐ Version control (Git) integrated with Talend Studio for all job development
- ☐ Deployment promotion (Dev → Test → Prod) follows a documented approval gate

7. Role and Access Configuration

Governance Role	Talend Role/Capability
Data Governance Office	Administrator
Technical Data Steward	Project Lead / Job Owner
Integration Developers	Job Developer (Dev/Test environments)
Operations	Job Conductor / Scheduler access (Prod)

8. Scheduling and Monitoring

- ☐ Centralize job scheduling through Talend Management Console (or equivalent orchestrator)
 - ☐ Configure job execution dashboards feeding into KPI Dashboard Templates
 - ☐ Configure SLA monitoring for time-sensitive jobs aligned to the Timeliness dimension in the Data Quality Standards
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9. Rollout Checklist

- ☐ Pilot domain jobs migrated/built and validated end-to-end
 - ☐ Lineage and quality results flowing into the catalog and reporting views
 - ☐ Technical Data Stewards trained on job standards and monitoring tooling
 - ☐ Production cutover scheduled per the Implementation Roadmap
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